

Problem-Solution Fit Canvas

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. Customer Segment(s) [CS] Residents and local volunteers in rainfall-prone regions; students/evaluators reviewing an end-to-end ML deployment.	6. Customer Limitations [CL] (e.g. budget, devices) Limited access to real-time hydrological sensors; rely on general weather apps; no ML background to interpret raw rainfall statistics.	5. Available Solutions Pros & Cons [AS] Generic weather apps (no flood-specific signal); government flood bulletins (infrequent, not localized); manual rule-of-thumb rainfall thresholds.
2. Problems / Pains + its Frequency [PR] No fast way to translate raw weather readings into a flood/no-flood signal; recurring every monsoon season.	9. Problem Root / Cause [RC] Flood risk assessment traditionally requires domain expertise or manual threshold lookups; no accessible tool bridges rainfall data and a plain-language risk read.	7. Behavior + its Intensity [BE] Users check weather conditions daily during storms (high frequency, high anxiety during active rainfall events).
3. Triggers to Act [TR] Heavy rainfall forecast, visible waterlogging nearby, official weather advisories.	10. Your Solution [SL] A Flask web app where users enter 10 weather/rainfall readings and instantly receive a flood/no-flood prediction with a visual risk gauge, from a model trained on historical rainfall data.	8. Channels of Behavior [CH] Online: Web browser access on desktop/mobile. Offline: Word-of-mouth sharing of the tool's link during community discussions.
4. Emotions Before / After [EM] Before: anxious, uncertain about local risk. After: informed and able to act early, or reassured when risk is low.		